InSAR Suite

User Guide — Versions 3.2.8 (QGIS 3) / 3.3.4 (QGIS 4)

Integrated suite for PSI data analysis in QGIS

Giovanni Montini

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1. Introduction to the plugin

InSAR Suite is an open-source QGIS plugin that brings together, in a single toolbar, a set of tools for the analysis of PSI (Persistent Scatterer Interferometric SAR) data. Its aim is to streamline the PSI data analysis workflow by providing, within a single interface, tools for data loading, geometric decomposition, visibility analysis and statistical analysis of time series.

The plugin is designed for local analyses supporting landslide and subsidence hazard assessment, not for the analysis of entire regional-scale datasets. The results it produces must always be evaluated together with other baseline data (geological surveys, hydrological data, geomorphological studies) and verified in the field, since they are a supporting element for hazard and risk analysis, not a standalone and conclusive assessment.

The Suite includes the following six modules:

Parameter	Description
InSAR Load	Loads PSI layers from file (GeoPackage, Shapefile, GDB) through a polygonal index layer (quadro di unione), or reactivates an index layer already present in the project.
InSAR EWUD	Reconstructs the velocity vector in the East-West / Up-Down plane from the LOS velocities of ascending and descending orbits, with automatic creation of the resampling grid.
InSAR VIS	Computes the percentage of detectable movement (pc_mov) as a function of the SAR geometry and of the terrain morphology derived from a DEM.
InSAR TS	Statistical analysis of PSI time series: data quality, automatic kinematic analysis, STL decomposition, piecewise non-linear analysis, temporal anomaly detection, comparison between zones.
InSAR Polygons	Automatic delineation of deformation areas from PS: classification by velocity threshold, geometric buffer and dissolve by class, validation and morphological smoothing of the boundary. Polygon output with statistical attributes (mean velocity, no. of PS, area).
InSAR Report	Automatically generates a summary report of the PSI analysis over a study area (drawn, loaded from file or selected from a layer already in the project): number of PS, density, areal coverage, kinematic coherence, visibility, velocity and acceleration (with standard deviation) for the ascending and descending datasets, and mean EWUD vector. Can be exported to PNG or CSV.

2. Installation

2.1 Requirements

- QGIS 3.16 – 3.99 (plugin v3.2.x) or QGIS 4.00+ (plugin v3.3.x)
- Python 3 with the libraries: pandas, numpy, matplotlib, scipy, statsmodels, pyproj, mplcursors
- Standard QGIS libraries: processing, qgis.core, qgis.gui, qgis.PyQt

2.2 Installation from the QGIS Repository (recommended)

1. Open QGIS and go to Plugins → Manage and Install Plugins → All.
2. Type InSAR Suite in the search bar.
3. Select the plugin from the list and click Install Plugin.
4. When finished, check that the plugin is active in the Installed tab.

ℹ New versions are published in the official repository after review by a moderator (usually within one working day). To be notified of updates once they are approved, enable Check for updates on startup in Plugins → Manage and Install Plugins → Settings.

2.3 Installation from ZIP

5. Open QGIS and go to Plugins → Manage and Install Plugins → Install from ZIP.
6. Select the ZIP file that matches your QGIS version: InSAR_Suite_v3.2.8_QGIS3.zip for QGIS 3, InSAR_Suite_v3.3.4_QGIS4.zip for QGIS 4.
7. Click Install Plugin.

2.4 Manual installation

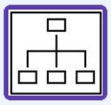
Copy the InSAR_Suite/ folder into the QGIS plugins directory:

- Windows (QGIS 3): C:\Users\<user>\AppData\Roaming\QGIS\QGIS3\profiles\default\python\plugins\
- Windows (QGIS 4): C:\Users\<user>\AppData\Roaming\QGIS\QGIS4\profiles\default\python\plugins\
- Linux / macOS (QGIS 3): ~/.local/share/QGIS/QGIS3/profiles/default/python/plugins/
- Linux / macOS (QGIS 4): ~/.local/share/QGIS/QGIS4/profiles/default/python/plugins/

3. Interface — The InSAR Suite Toolbar

On startup the plugin adds a toolbar named InSAR Suite with 12 buttons organized in six sections (LOAD, EWUD, VIS, TS, POLYG, REPORT), separated by blue vertical dividers. All buttons are also accessible from the Plugins → InSAR Suite menu.

Section	Button	Function	
LOAD		**Carica PS da File**	Opens the index layer (quadro di unione) selection window. Supports GeoPackage, Shapefile, Geodatabase.
LOAD		**Riattiva Quadro di Unione**	Reconnects the selection signal to an index layer already present in the project.
EWUD		**InSAR EWUD**	East-West / Up-Down decomposition from ascending/descending pairs.
VIS		**InSAR VIS**	Computes the percentage of detectable movement (pc_mov) from SAR geometry and DEM.
TS		**TS – Qualità del dato**	Histogram + normal curve, Q-Q plot, boxplot, robust statistics (including skewness and kurtosis), Shapiro-Wilk. Transformation selector. “Carica PS coerenti in QGIS” button available in the top bar.
TS		**TS – Analisi automatica**	Mean series ±1σ, linear trend, interactive tooltip. Loads the table into QGIS.

Section	Button	Function	
TS		**TS – Scomposizione STL**	Trend / seasonality / residual decomposition.
TS		**TS – Analisi non lineare**	Piecewise regression, BIC optimization, segment summary table.
TS		**TS – Anomalie temporali**	Detection of anomalous acquisitions from residuals and consecutive changes. ⚠ ANOMALIA tooltip.
TS		**TS – Confronto tra zone**	Comparison of mean series between 2–3 zones. Non-modal panel.
POLYG		**InSAR Polygons**	Automatic delineation of deformation areas by geometric buffer and dissolve on unstable PS.
REPORT		**InSAR Report**	Generates the PSI analysis summary report (N PS, density, areal coverage, kinematic coherence, visibility, velocity and acceleration) over a study area. Can be exported to PNG or CSV.

4. InSAR Load module

The Load module handles the loading of point PSI layers through a mechanism based on a polygonal index layer (quadro di unione): a polygon layer in which each feature stores, in one of its attributes, the name of the corresponding PS layer to be loaded. Selecting one or more polygons on the map automatically loads the associated PS layers, giving fast and spatially controlled access to large datasets organized by sheets or tiles.

The module supports the following formats: GeoPackage (.gpkg), Shapefile (.shp), ESRI File Geodatabase (.gdb).

4.1 Carica PS da File (Load PS from File)

Loads the index layer directly from a file on disk.

Procedure:

8. Click the Carica PS da File button in the toolbar.
9. Select the format of the source file: GeoPackage (.gpkg), Shapefile (.shp) or Geodatabase (.gdb).
10. Select the file in the file browser window.
11. If the file contains more than one polygon layer, choose the index layer from the list shown.
12. Choose the index layer field that contains the names of the PS layers.
13. The index layer is added to the project (with visibility turned off). Select one or more polygons on the map to load the corresponding PS layers.

If the shapefile format is selected, the PS point layers and the polygonal index layer must be located in the same folder.

ℹ The index layer is added with visibility turned off so as not to hide the base map, while the PS layers are loaded with visibility turned on.

4.2 Riattiva Quadro di Unione (Reactivate Index Layer)

Reactivates the selection mechanism on an index layer already present in the QGIS project, without reopening the file on disk. Useful when reopening a project that has already been set up.

Procedure:

14. Click the Riattiva Quadro di Unione button in the toolbar.
15. Select from the list the polygon layer to be used as the index layer.
16. Choose the field containing the names of the PS layers.
17. Select the polygons on the map to load the corresponding layers.

5. InSAR EWUD module

The EWUD module reconstructs the deformation velocity vector in the East-West / Up-Down plane from the velocities measured along the line of sight (LOS) in the two acquisition geometries, ascending and descending. Since each geometry "sees" the movement from a different direction, combining the two LOS measurements makes it possible to solve the system of geometric equations and derive the two independent components of the vector: the horizontal East-West component (E) and the vertical Up-Down component (U). The velocity magnitude in the EW-UD plane and the prevailing direction of movement are then derived from the (E, U) vector. The North component is conventionally neglected because SAR satellites in near-polar orbits are poorly sensitive to movements in the North-South direction, so that component is poorly constrained. The 2×2 system is:

$$Va = Ca_e \cdot E + Ca_u \cdot U$$
$$Vd = Cd_e \cdot E + Cd_u \cdot U$$

where the geometric coefficients Ca_e = sgn·sin(θ_a)·cos(φ_a), Ca_u = cos(θ_a), Cd_e = sgn·sin(θ_d)·cos(φ_d), Cd_u = cos(θ_d) depend on the off-nadir (θ) and orientation (φ) angles of the satellite for the ascending (a) and descending (d) orbits. The factor sgn = −1 for a right-looking satellite. The qualitative classification of the velocity field is computed from the angle of the (E, U) vector in the EW-UD plane (Fig. 1).

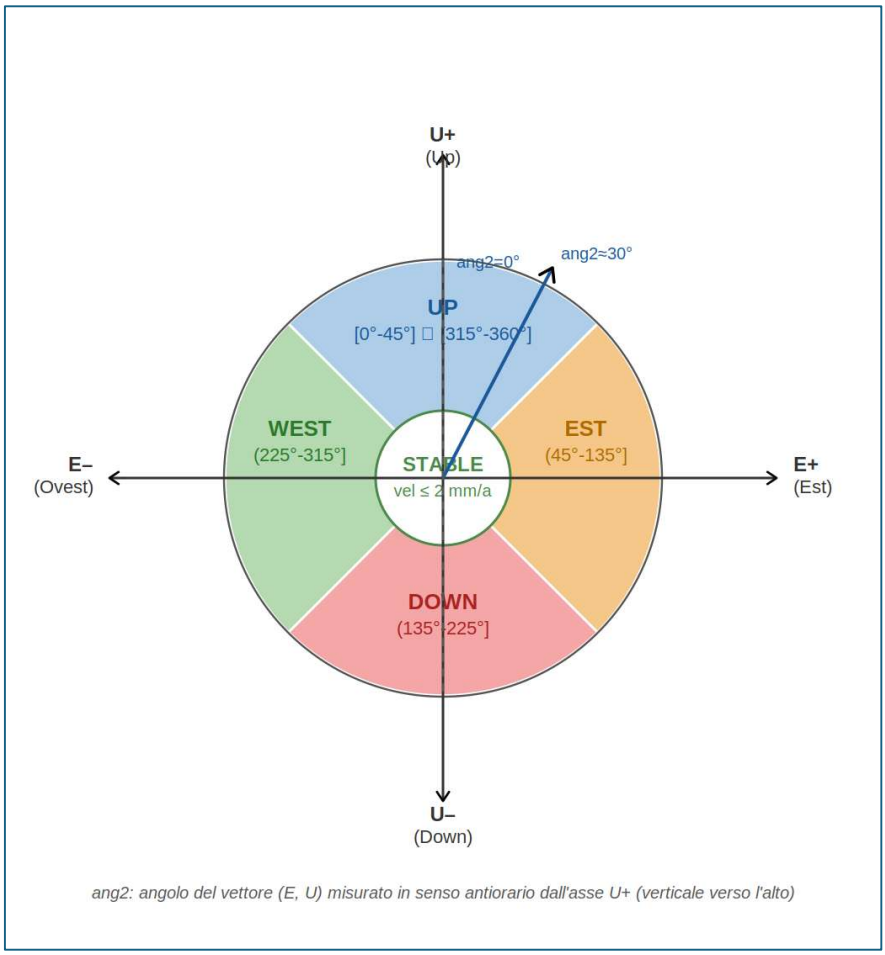


Fig. 1 — Scheme of the vprev classification as a function of the angle ang2 of the velocity vector in the EW-UD plane. The angle ang2 is measured counterclockwise from the U+ axis (vertical, pointing up); the STABLE zone corresponds to velocities vel ≤ 2 mm/yr regardless of direction.

To decompose the values recorded in the two orbits and reconstruct the mean deformation velocity vector in the EW-UD plane, a resampling grid must first be defined, so that each cell is assigned a single mean velocity value for each of the two acquisition geometries; the EWUD vector can then be reconstructed. The module therefore consists of two tabs.

5.1 Tab ① — Crea Griglia PS (Create PS Grid)

Generates a regular polygon grid over the area of interest, keeping only the cells that contain at least one ascending and one descending PS point. The parameters required in this tab are the following:

Parameter	Description
Estensione griglia	Area of interest. It can be drawn on the map, taken from a layer or entered manually.
Lato cella	Cell side length in meters (default: 50 m). Typical values: 50–200 m depending on PS density.
Tipo cella	Cell shape: Quadrata (square, default) or Esagonale (hexagonal). Hexagonal cells are preferable for areas with an irregular PS distribution.
PS Ascendenti	Point layer of PS in ascending orbit.
PS Discendenti	Point layer of PS in descending orbit.
Output PS Grid	Output file path. If left empty, a temporary layer is created.

If a resampling grid is already available, this tab can be skipped and the grid used directly in Tab ②.

5.2 Tab ② — Decomposizione EWUD (EWUD Decomposition)

For each grid cell, computes the mean ascending and descending velocities and derives the E and U components by solving the linear system. The parameters required in this tab are the following:

Parameter	Description
Griglia di ricampionamento	Polygon layer created in Tab ①.
Satellite / missione	Preset menu: selecting a satellite pre-fills the heading and off-nadir values.
PS Ascending – Layer / Velocità	Point layer and ascending LOS velocity field (mm/yr).
Azimut (track angle) Ascending	Orientation angle of the satellite with respect to North. Typical S1 ASC: 349°.
Off-nadir Ascending	Angle between the local vertical and the LOS. Typical S1: 20°–46°.
PS Descending – Layer / Velocità	Point layer and descending LOS velocity field (mm/yr).
Azimut / Off-nadir Descending	Geometry in descending orbit. Typical S1 DESC: 191°.
Right-looking	Checked for standard satellites. Uncheck only for left-looking geometries.
Centroidi EWUD / Poligoni EWUD	Output point and polygon layers with EWUD values. If left empty, two temporary layers are created.

5.3 Outputs

The EWUD module produces two vector layers: a polygon layer (Poligoni_EWUD) and a point layer of centroids (Centroidi_EWUD), both with an automatically applied color legend. The main fields are the East-West component (E, mm/yr), the vertical Up-Down component (U, mm/yr), the velocity magnitude of the deformation vector in the EW-UD plane (vel, mm/yr), the angle of the deformation vector in the EW-UD plane (ang2, degrees) and the qualitative classification (vprev: STABLE, UP, DOWN, EST, WEST, Fig. 1). The number (Na and Nd) and mean velocity (Va and Vd) of the ascending and descending PS falling within each cell are also reported.

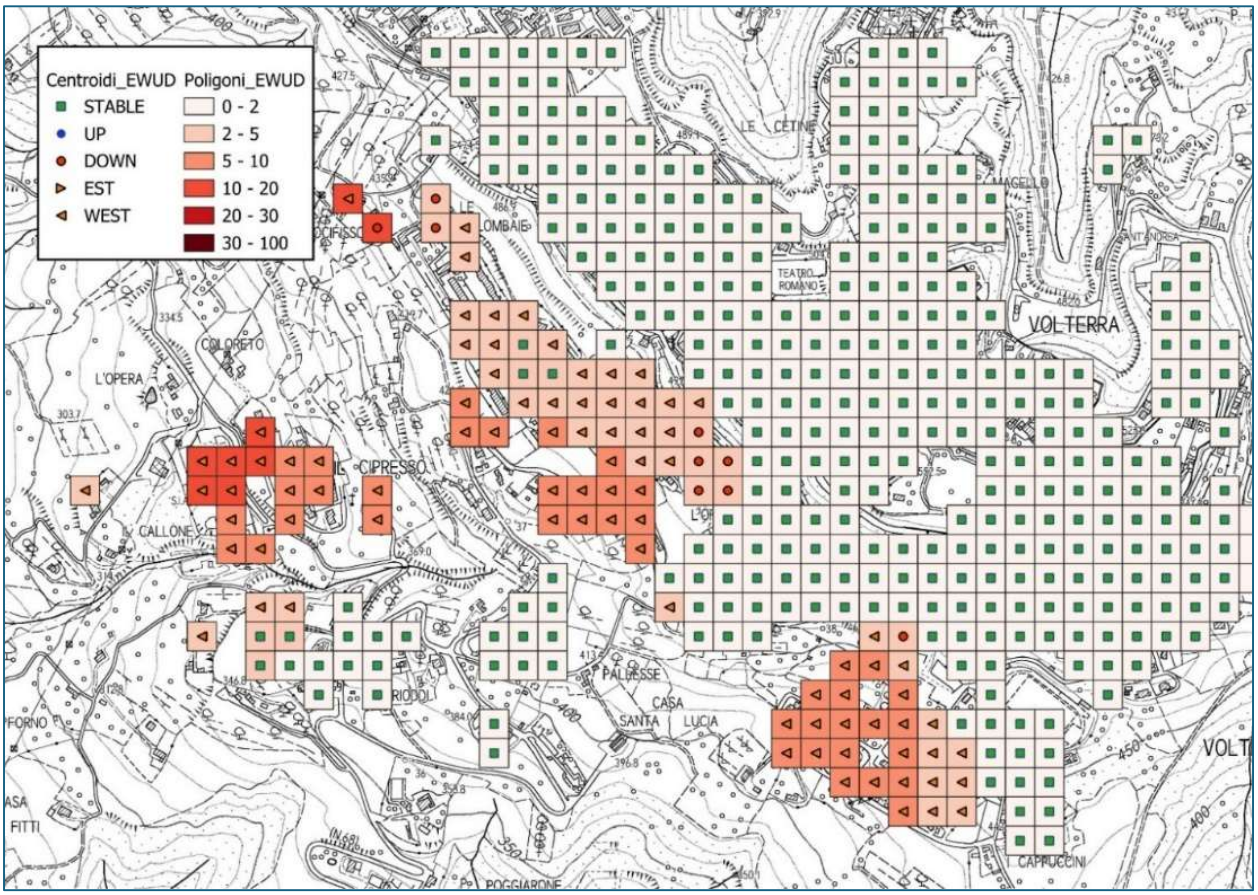


Fig. 2 — Centroidi_EWUD and Poligoni_EWUD layers on a QGIS map: the centroids show the prevailing direction of movement in the EWUD plane, and the polygons are color-coded according to their velocity in the EWUD plane, expressed in mm/yr.

Combining the two layers lets the user read, at the same time, the spatial distribution of the intensity and of the prevailing direction of movement: cells with a more intense color in the polygon layer indicate high velocities in the EW-UD plane, while the overlaid point symbols show their prevailing direction (Fig. 2).

5.4 Available satellite presets

The satellite presets contain average values for central-northern Italy, but these may vary from area to area. The characteristic angular values (azimuth and off-nadir) can therefore be customized according to the area under study and the information given in the metadata.

Parameter	Description
Sentinel-1 (EGMS - Italia centro-settentrionale)	ASC: Azimuth 349°, off-nadir 42° | DESC: Azimuth 191°, off-nadir 38°
Sentinel-1 (generico)	ASC: Azimuth 348°, off-nadir 33° | DESC: Azimuth 192°, off-nadir 33°
ERS / Envisat	ASC: Azimuth 347°, off-nadir 23° | DESC: Azimuth 193°, off-nadir 23°
ALOS / ALOS-2	ASC: Azimuth 350°, off-nadir 34° | DESC: Azimuth 190°, off-nadir 34°
COSMO-SkyMed	ASC: Azimuth 345°, off-nadir 30° | DESC: Azimuth 195°, off-nadir 30°
TerraSAR-X / TanDEM-X	ASC: Azimuth 350°, off-nadir 35° | DESC: Azimuth 190°, off-nadir 35°
RADARSAT-2	ASC: Azimuth 350°, off-nadir 35° | DESC: Azimuth 190°, off-nadir 35°
Personalizzato	All angular fields become manually editable.

6. InSAR VIS module

For each PS point, the VIS module computes the percentage of gravitational movement detectable by the SAR acquisition geometry. The underlying assumption is that the actual displacement occurs along the line of steepest slope. The percentage of detectable movement (pc_mov) is defined as:

$$pc_mov = |\ LOS \cdot U\ | \times 100 = |\ \cos(\alpha)\cdot\sin(\theta)\cdot\sin(\beta-\varphi) + \sin(\alpha)\cdot\cos(\theta)\ | \times 100$$

where α is the terrain slope, β is the aspect, θ is the off-nadir angle and φ is the satellite track angle. For flat terrain (slope ≤ 3°) the calculation reduces to: pc_mov = cos(θ) × 100.

pc_mov values close to 100% indicate that the SAR geometry is optimal for detecting the expected movement, while low values (< 20-30%) indicate that the movement is largely invisible to the sensor. This analysis is particularly useful in landslide areas, where the orientation of the slope relative to the acquisition geometry can strongly affect the detectability of the signal.

6.1 Input parameters

Parameter	Description
Shapefile PS (punti)	Point layer of the PSI.
Raster quote (DEM)	Digital terrain model.
Estensione elaborazione	Area of interest.
Cella ricampionamento DEM (m)	DEM resampling resolution (default: 100 m).
Orbita	Ascending (ASC) or Descending (DESC).
Satellite	Preset menu with pre-filled parameters.
Azimut / Off-nadir	Orientation angle (expressed in positive degrees, e.g. 349° for S1 ASC EGMS - central-northern Italy) and incidence angle of the satellite.
Nome layer output	Name of the in-memory layer (default: calcolo_pc_rilevata).

6.2 Output — Fields added to the PS layer

The output file keeps all the fields of the input file, with the following three fields added:

Parameter	Description
esp1	Terrain aspect at the PS point (degrees, 0=North, clockwise).
inc1	Terrain slope at the PS point (sexagesimal degrees).
pc_mov	Percentage of detectable movement (0–100%). High values = good SAR sensitivity to the expected movement.

The output layer is temporary by default. It can be saved permanently by checking the “Salva su file (GeoPackage)” box in the VIS module window and specifying the path. The layer has an automatically applied legend with four classes of the percentage of movement detected by the satellite: 0-25%, 25-50%, 50-75% and 75-100%. An example of the graphical output of the VIS module is shown in Fig. 3, where the different percentages of movement detected by the PS of the ascending dataset are displayed over the digital terrain model, which highlights the spatial variability of the morphometric parameters in the study area. It should nevertheless be kept in mind that the percentage of movement is computed assuming an actual movement along the line of steepest slope, but this is not always the case, for example for DSGSDs (deep-seated gravitational slope deformations).

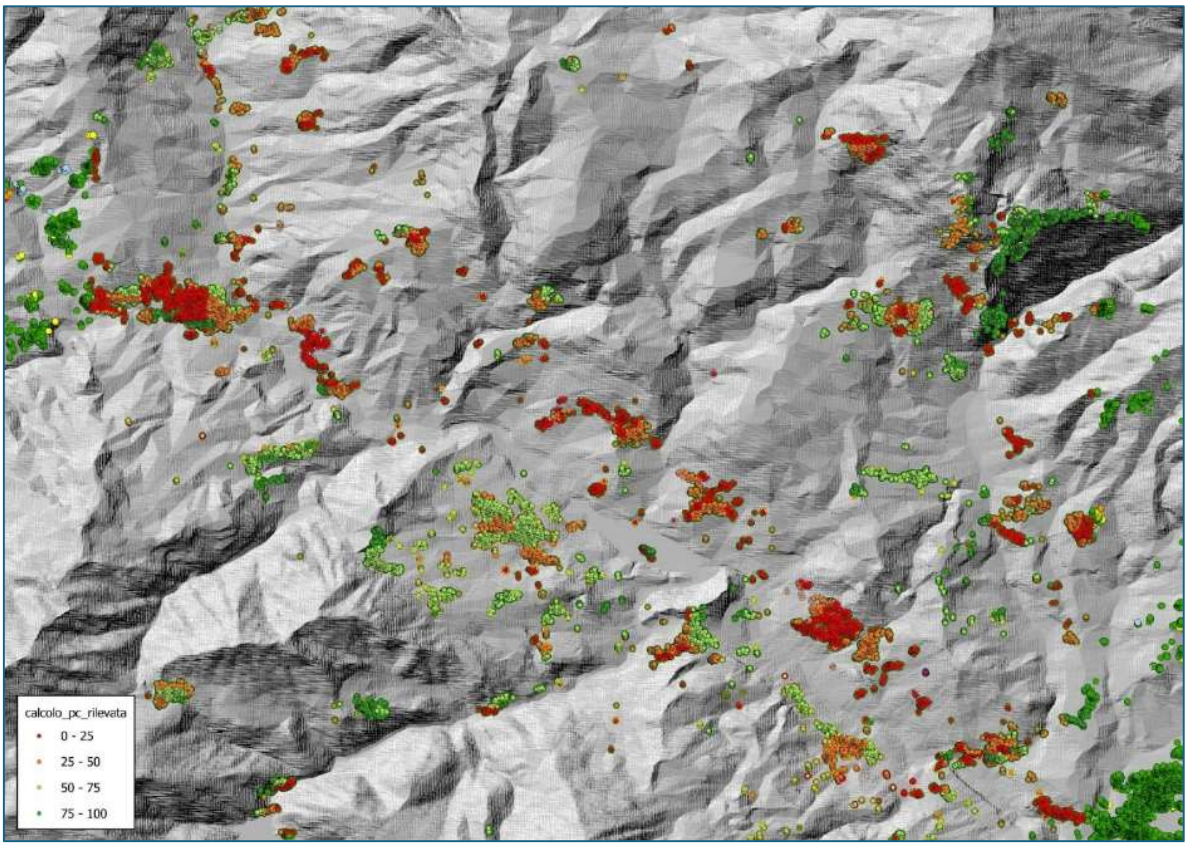


Fig. 3 — Map of the percentage of movement detected by the satellite (pc_mov) on an ascending PS layer: color scale from red (low visibility, movement nearly perpendicular to the LOS) to green (high visibility, movement nearly parallel to the LOS). Hillshaded DEM in the background.

⚠ For terrain with slope ≤ 3° (flat terrain) the aspect is not geometrically meaningful: the calculation automatically assumes purely vertical movement, with pc_mov = cos(off-nadir) × 100.

7. InSAR TS module

The TS module brings together six tools for the statistical analysis of PSI time series. They all work on the active point vector layer in QGIS and on the features manually selected on the map. For each analysis in this module, the user is first asked to define a correlation threshold based on the Pearson correlation matrix between the time series of the selected PS: a PS is considered coherent if its correlation with at least half of the other PS in the group exceeds a user-configurable threshold (default: r ≥ 0.85). This makes it possible to analyze only the PS that show a similar kinematic behavior among all those selected, or to examine all of them by setting the correlation threshold to zero.

7.1 Prerequisites

- Set the PS point layer as the active layer (click the layer in the QGIS Layers panel — the layer is highlighted in blue).
- Select one or more PS points on the map using the standard QGIS selection tools.
- Check that the layer contains displacement fields in the DYYYYMMDD format (e.g. D20170101) or YYYYMMDD format (e.g. 20170101), i.e. one field for each SAR acquisition date. The plugin automatically recognizes both formats.

⚠ If, when a TS button is clicked, no PS layer is active or no points are selected, the plugin displays an explanatory warning window. If the correlation threshold finds no coherent PS, the "Nessun PS coerente trovato!" window is shown with suggestions for the user.

7.2 TS — Qualità del dato (Data Quality)

Analyzes the quality and homogeneity of the selected PS, producing three side-by-side plots and a statistics panel. The initial dialog asks for the correlation threshold (default: 0.85). Once the plot window is open, a data transformation can be chosen and plots and statistics are recalculated in real time.



Fig. 4 — Plot window of the Qualità del dato tool: frequency histogram of the velocities with the theoretical normal curve N(μ, σ) overlaid, vertical lines for the mean (solid green), median (dashed orange) and ±1σ (dotted green); Q-Q plot with reference line and R²; boxplot with individual data points overlaid and an explanatory legend below the plot. The statistics panel reports n PS, mean, standard deviation, median, IQR, MAD, relative dispersion, outliers (robust z-score > 2.5) and the Shapiro-Wilk test. The top bar lets the user choose the transformation (Nessuna / Logaritmica / Yeo-Johnson / Box-Cox) and recalculate all plots and statistics in real time.

A highly skewed distribution or a large number of outliers may indicate PS with anomalous behavior that should be checked before proceeding with further analyses. The top bar of the window includes the “Carica PS coerenti in QGIS” button, which creates a temporary layer containing only the coherent PS, with all the original attributes of the source PS layer.

Plots produced:

Parameter	Description
Histogram + normal curve	Frequency distribution with N(μ,σ) curve. Outlier bars in red. Vertical lines for median (orange), mean (green) and ±1σ.
Q-Q Plot	Quantile-Quantile plot with R². Outlier points highlighted in red.
Boxplot + individual data	Boxplot with every PS overlaid. Green lines for mean ±1σ. Legend below the plot.
Statistics panel	n PS, mean, std. deviation, median, IQR, MAD, skewness, kurtosis (excess), relative dispersion, outliers (z-rob > 2.5), Shapiro-Wilk.

Available transformations:

Parameter	Description
Nessuna	Analysis on the original velocity values (no transformation).
Logaritmica	ln(v − min + ε). Useful for positively skewed distributions.
Yeo-Johnson	Optimized transformation that also works with negative values. Recommended for PSI data.
Box-Cox	Classic transformation; requires positive values (automatic shift applied).

7.3 TS — Analisi automatica (Automatic Analysis)

Reconstructs the mean displacement time series for the group of selected PS and estimates the linear velocity rate. It computes the correlation between the series of the individual PS (configurable threshold) and produces:

- A plot with the mean series, ±1σ band, linear trend with velocity and R², and an interactive tooltip on each acquisition.
- A “Carica tabella in QGIS” button in the plot toolbar, which loads the Serie_media_PS_coerenti table layer, which can be saved from the Layers panel. A “Carica PS coerenti in QGIS” button creates a temporary layer containing only the coherent PS, with all the original attributes.

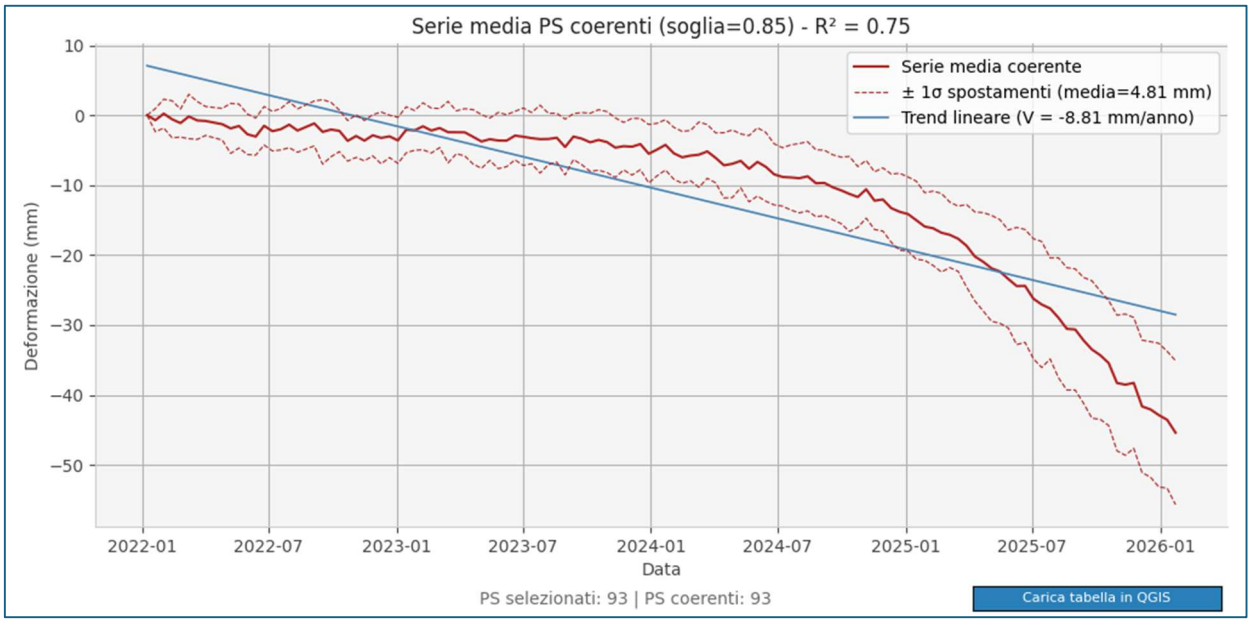


Fig. 5 — Mean time series (blue line) with ±1σ band (gray area) and linear regression line (dashed red line). The estimated velocity and R² are shown in the legend. Each point of the series is interactive: hovering the mouse shows the date and value.

ℹ If only one PS is selected, the correlation threshold is ignored and the series of that single point is analyzed.

7.4 TS — Scomposizione STL (STL Decomposition)

Decomposes the mean time series into the trend T(t), seasonal S(t) and residual R(t) components using the STL algorithm (Seasonal and Trend decomposition using Loess). The seasonal component in PSI data may be of thermal origin (expansion of metal targets) or hydrological origin (changes in soil moisture), with a typical magnitude of 1-5 mm/yr for Sentinel-1.

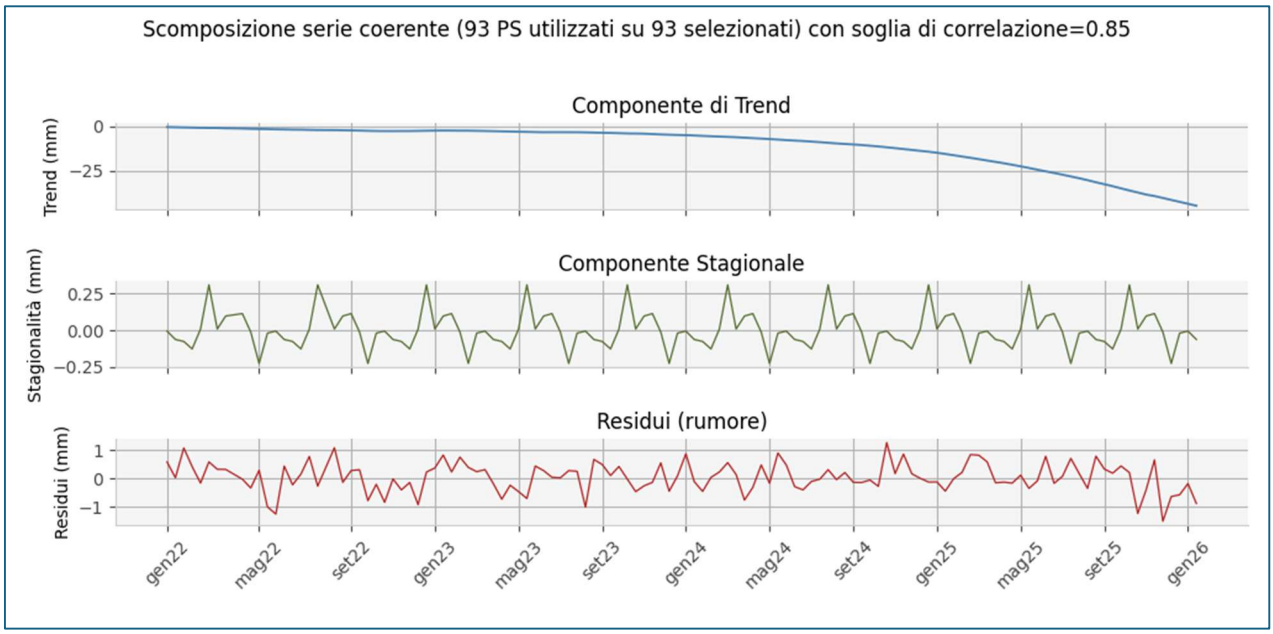
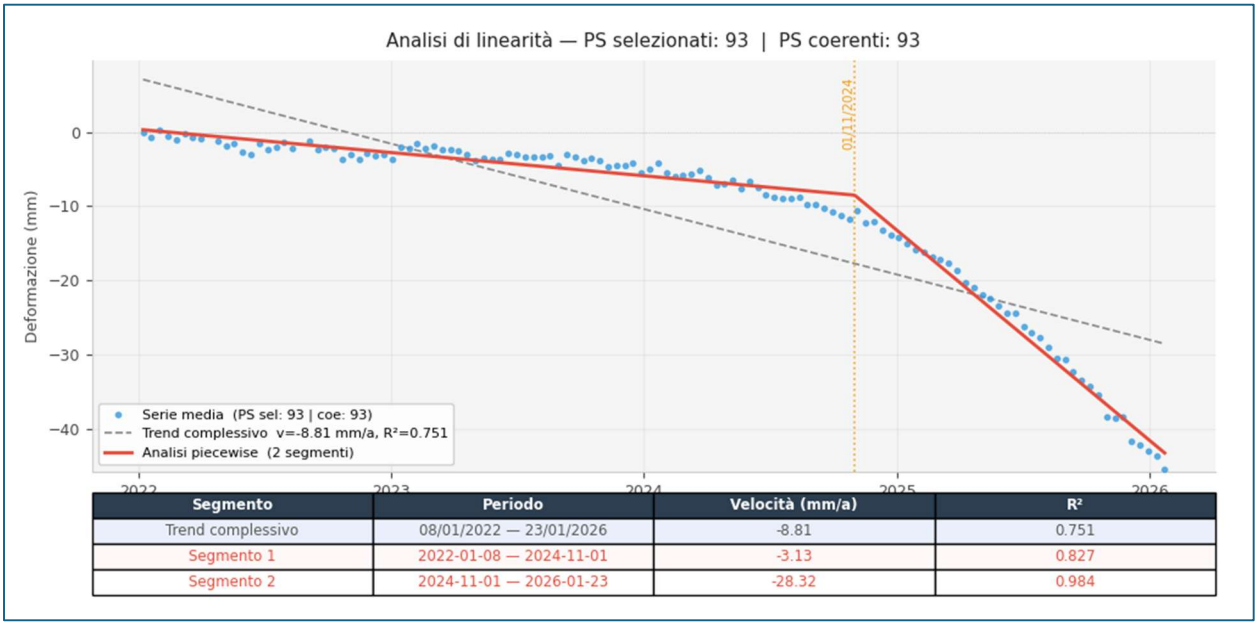


Fig. 6 — STL decomposition of the mean series: from top to bottom, trend component T(t), seasonal component S(t) and residual R(t). The magnitude of the seasonal component relative to the trend indicates the weight of annual oscillations on the deformation signal.

7.5 TS — Analisi non lineare piecewise (Piecewise Non-linear Analysis)

Assesses the linearity of the time series by piecewise regression with hinge functions (no external dependencies). The dialog asks for two parameters: the correlation threshold for the coherence filter and the maximum number of segments to test (2–5, default 2). The BIC automatically identifies the optimal number within the chosen range; limiting the range reduces computation time when only a few breakpoints are expected.

The breakpoints identified correspond to moments when the deformation rate changes significantly, and may be associated with physical events (heavy rainfall, earthquakes, seasonal hydrological changes).



Segmento	Periodo	Velocità (mm/a)	R²
Trend complessivo	08/01/2022 — 23/01/2026	-8.81	0.751
Segmento 1	2022-01-08 — 2024-11-01	-3.13	0.827
Segmento 2	2024-11-01 — 2026-01-23	-28.32	0.984

Fig. 7 — Piecewise analysis: mean series (blue dots) with overall trend (dashed gray) and piecewise linear model (red). Breakpoints are shown by vertical orange lines with the corresponding date. The summary table at the bottom reports period, velocity and R^2 for each segment.

ℹ Choose a small maximum number (e.g. 3) if at most one breakpoint is expected: the calculation will be considerably faster.

7.6 TS — Anomalie temporali (Temporal Anomalies)

Identifies anomalous acquisition dates in the mean series of the coherent PS. An acquisition is flagged as anomalous if at least one of the following conditions is met:

- Its residual from the linear trend exceeds a configurable threshold (default: 3σ of the residuals).
- The change from the previous acquisition exceeds a configurable threshold (default: 5 mm).

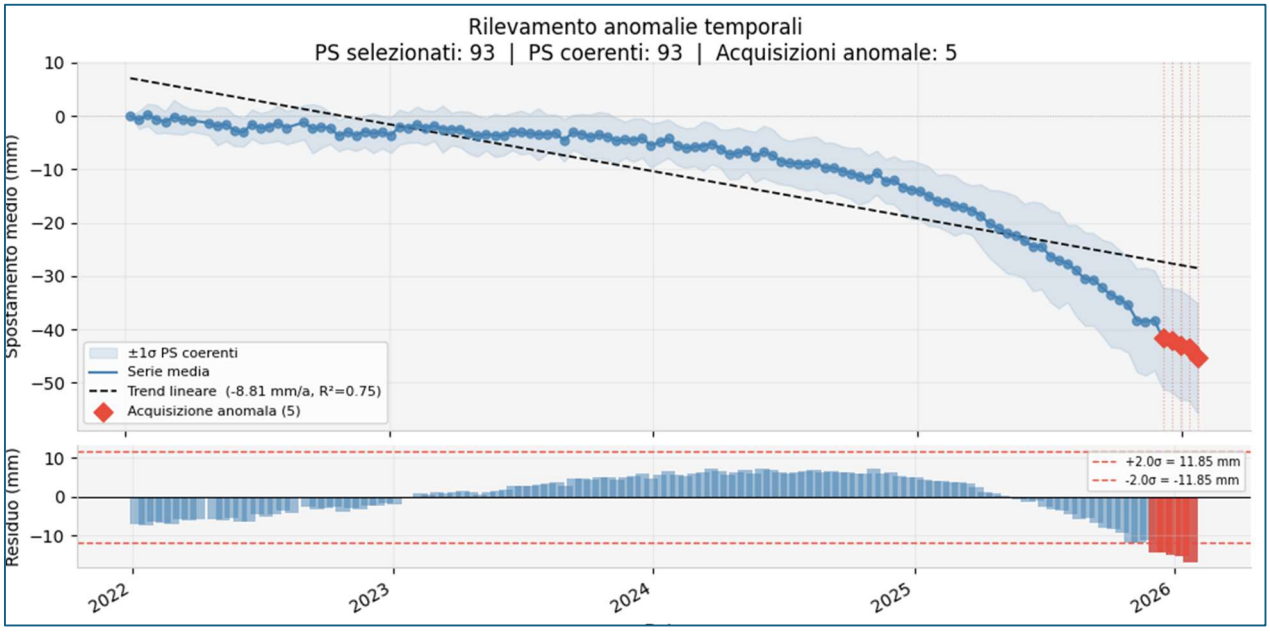


Fig. 8 — Temporal anomaly detection: mean series with anomalous acquisitions (red diamonds) and dashed red vertical lines. The lower plot shows the residuals with the alert thresholds. The interactive tooltip shows date and displacement, with the ⚠ ANOMALIA tag for out-of-threshold acquisitions.

The interactive tooltip on the mean series shows date, displacement and, if the acquisition is anomalous, the ⚠ ANOMALIA tag on a light red background. The breakpoints from the piecewise analysis can be compared with the anomalous dates to assess whether regime changes coincide with lower-quality acquisitions.

7.7 TS — Confronto tra zone (Zone Comparison)

Compares the mean time series of two or three spatially distinct PS zones. The control panel is non-modal: it stays open and on top while the user selects points on the map, without blocking interaction with QGIS.

Workflow:

1. Select the PS of Zone A on the map → click Conferma Zona A in the panel.
2. Select the PS of Zone B → click Conferma Zona B.
3. (Optional) Select the PS of Zone C → click Conferma Zona C.
4. Click Calcola confronto.

An example of the graphical output obtained by comparing PS located in three spatially distinct areas is shown in Fig. 9.

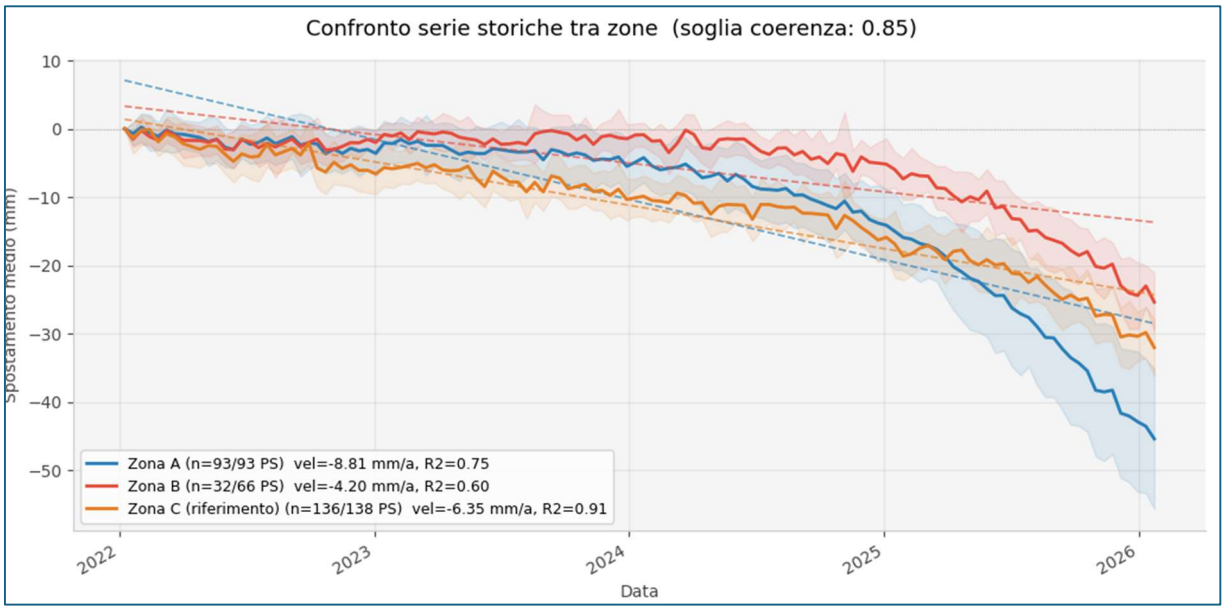


Fig. 9 — Zone comparison: mean time series of three PS zones overlaid, with ±1σ bands (semi-transparent areas) and OLS regression lines (dashed lines). Colors distinguish the zones: blue (A), red (B), orange (C). The title shows the coherence threshold used.

Reading several spatially separate zones at once makes it possible to assess whether adjacent areas show different kinematic behaviors, supporting the geological and geomorphological interpretation of the dataset.

8. InSAR Polygons module

The InSAR Polygons module automatically delineates deformation areas from a PS vector layer. The processing workflow is based on geometric buffer and dissolve, and reproduces the manual photo-interpretation procedure in an automatic and repeatable way.

8.1 Input parameters

The module window requires: PS point layer (ascending or descending), velocity column (mm/yr), point selection mode (selected / canvas / all), unstable velocity threshold (default 2 mm/yr), buffer and search radius in m (default 50 m), minimum number of unstable PS in the neighborhood (default 5), minimum unstable/total ratio 0-1 (default 0.75), minimum number of PS in the final polygon (default 5), optional morphological smoothing.

8.2 Processing workflow

Processing runs in a non-blocking QgsTask. The workflow consists of: (i) classification of the unstable PS into six velocity classes (< -10, -10/-5, -5/-2, +2/+5, +5/+10, > +10 mm/yr); (ii) creation of circular buffers of radius R around each unstable PS that meets the validation criteria; (iii) dissolve by class with hierarchical clipping to avoid overlaps; (iv) final validation by number of PS and unstable ratio; (v) optional morphological smoothing. The log in the window shows the details in real time.

8.3 Output

The module produces a polygon vector layer (Fig. 10) with the attributes: cluster_id, vel_class, priority, n_ps, n_unstable, vel_mean, vel_std, vel_min, vel_max (mm/yr), area_km2. A legend classified by velocity class is applied automatically.

> **i** Starting from version 3.3.0 (QGIS 4), the output can also be saved permanently to GeoPackage by checking the "Salva su file (GeoPackage)" box in the module window — a feature not available in version 3.2.x for QGIS 3.

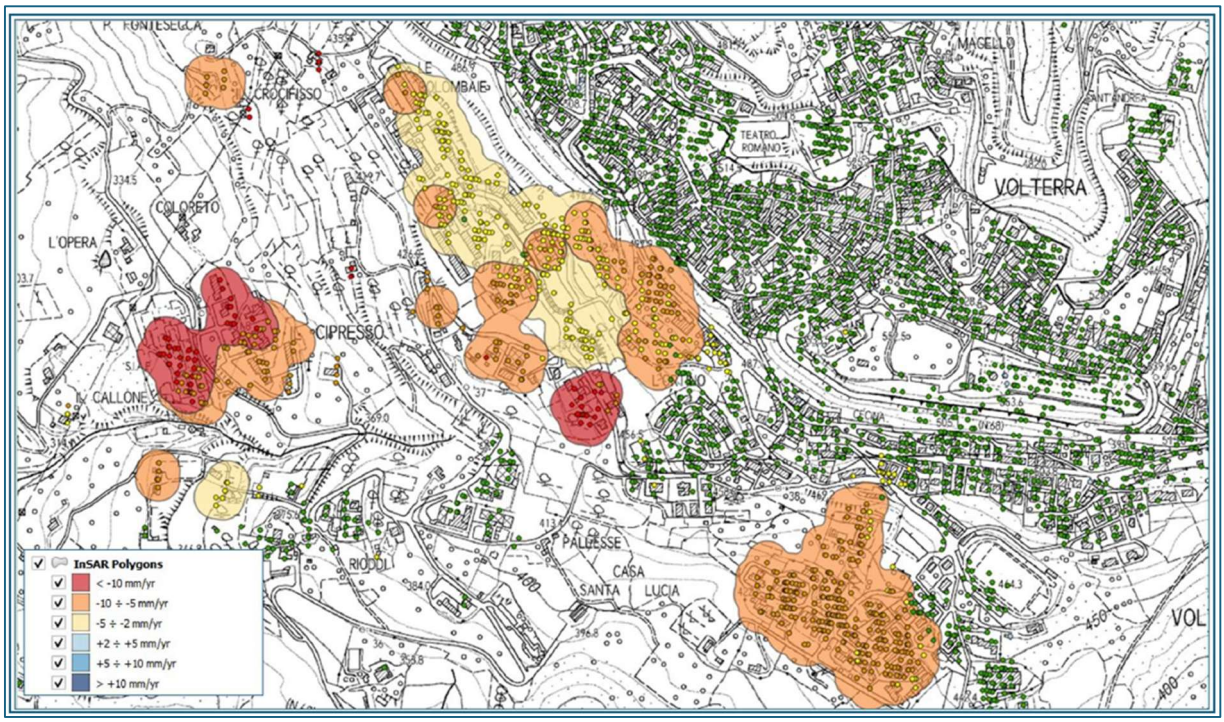


Fig. 10 — Delineation of deforming areas: analysis of the descending point dataset, with creation of the polygons that best approximate the deforming areas according to the criteria configurable in the InSAR Polygons module. The areas thus defined are classified according to the mean velocities of the points falling within them.

9. InSAR Report module

The InSAR Report module automatically generates a summary report of the PSI analysis over a study area (typically a landslide), intended as a rough estimate for a working group. It summarizes in a single window the data reliability indicators (number of PS, density, areal coverage, kinematic coherence) together with the kinematic values (velocity, acceleration, EWUD vector), separately for the ascending and descending datasets.

9.1 Study area

The study area can be defined in three equivalent ways: drawn by hand on the map (left-click to add vertices, right-click or double-click to close the polygon), selected from a polygon layer already present in the project via the drop-down menu, or loaded from an external vector file (GeoPackage or Shapefile) with the "… da file" button. If the file or layer contains several polygons, they are merged into a single area.

9.2 Input parameters

The module window requires the following parameters:

Parameter	Description
Area di studio	Polygon defined by drawing by hand on the map, selecting a layer already in the project, or loading an external vector file (§9.1).
PS Ascendente + campo velocità	PS point layer in ascending orbit and its LOS velocity field (mm/yr).
PS Discendente + campo velocità	PS point layer in descending orbit and its LOS velocity field (mm/yr).
Layer VIS Ascendente / Discendente	Optional. Layer with the pc_mov field already computed by the InSAR VIS module (§6).

Parameter	Description
Layer EWUD	Optional. Centroidi_EWUD or Poligoni_EWUD layer with the ang2, vel, E, U fields already computed (§5).
Raggio buffer	Radius (in meters) used to compute the areal coverage (default 50 m, see §9.3).
Soglia coerenza cinematica	Pearson correlation threshold for kinematic coherence (default 0.85, same meaning as in the TS module).
Soglia accelerazione	Minimum ratio between the velocity of the last segment and that of the previous one (2-segment piecewise analysis) above which an acceleration is flagged (default 1.5, see §9.6).

9.3 Areal coverage: note on the calculation method

The areal coverage is computed by creating a circular buffer (configurable radius, default 50 m) around each PS in the study area, geometrically merging all the buffers (dissolve) and comparing the resulting area with the total surface of the study area. Thanks to the geometric union, overlapping buffers (e.g. very closely spaced PS) are not counted more than once: the percentage always reflects the area actually covered, not the sum of the individual circles. The calculation is automatically performed in a metric (projected) coordinate system, even if the PS layer or the project use geographic coordinates (e.g. WGS84): in that case the plugin automatically determines the most appropriate UTM zone based on the centroid of the study area.

9.4 VIS and EWUD layers: no clipping required

> **i** The selected VIS and EWUD layers do not need to be clipped to the study area: the module automatically keeps only the features that fall geometrically within the area polygon and discards the rest. It is therefore normal and correct to select the "large" VIS and EWUD outputs, computed over an extent wider than the specific study area — no prior data preparation is needed.

9.5 Interpreting the results

PS density is expressed in PS/km². Areal coverage and kinematic coherence are percentages (0-100%); for kinematic coherence the correlation threshold used is also reported, with the same meaning as in the TS module. Mean visibility, velocity (overall and for coherent PS only) and EWUD velocity are reported in the "mean ± standard deviation" format, to give an immediate idea of the spread of the values as well, not just the mean value.
The EWUD angle is a circular quantity (a direction), so it cannot be averaged like an ordinary number: the module uses the vector mean (circular statistics), which is also correct for directions close to 0°/360°. The spread is expressed as the circular standard deviation; when the directions of the individual EWUD cells are too scattered for a meaningful standard deviation, the field shows "± N/D" instead of a misleading number. Hovering the mouse over the angle field also shows a tooltip with the value of the mean resultant length R (0-1): values close to 1 indicate a consistent direction, values close to 0 indicate directions scattered in all directions.

9.6 Acceleration detection

For each dataset (ascending and descending), the module computes an acceleration indicator by comparing the recent deformation velocity with the previous one. Unlike the "TS — Analisi non lineare piecewise" tool (§7.5), which is exploratory and lets the user freely choose the number of segments, here the model always uses exactly 2 fixed segments (a single breakpoint, whose optimal position is still searched for automatically): a meaningful velocity ratio always requires exactly two velocities to compare, and a variable number of segments would make the ratio itself meaningless.
The calculation is performed only on the PS that exceed the kinematic coherence threshold (the same selection used for "Velocità media coe", §9.5) — not on all the PS in the study area. Before the piecewise fit, the seasonal component (thermal or hydrological) is removed from the mean time series, following the same logic described for the TS module (§7.5): otherwise the detected

breakpoint could fall on a seasonal maximum or minimum instead of on a real change in the deformation rate.
The result is shown in the format "ratio× (month/year of the breakpoint)", for example "2.99× (05/2022)": the ⚠ symbol appears when the ratio exceeds the set threshold (default 1.5, configurable), and the ↑↓ symbol indicates a reversal of direction between the two segments (e.g. from subsidence to uplift). Hovering the mouse over the field shows a tooltip with the full details: velocities of the two segments, exact date of the breakpoint, threshold used. If there are fewer than 2 coherent PS, the data are insufficient (fewer than 6 acquisitions), or the velocity of the previous segment is too close to zero (numerically unstable ratio), the field shows "N/D".

> **i** The "TS — Analisi non lineare piecewise" tool (§7.5) now defaults to 2 segments (previously 5), but the user can still change this up to a maximum of 5. Unlike the Report module, that tool chooses the number of segments using the BIC, which may occasionally favor more segments than a geologist would expect: piecewise models with free (non-fixed) breakpoint positions make the BIC less conservative than expected, a known limitation of the method — not a calculation error. When in doubt, always check whether the additional breakpoints correspond to real changes in slope or to noise/residual oscillations.

9.7 Output

The "Genera report" button starts the calculation in the background (without blocking the QGIS interface) and, when finished, opens a window with the summary (Fig. 11). From there the report can be exported as a PNG image ("Esporta PNG" button) or as a CSV file with a field-value structure ("Esporta CSV" button, can be opened directly in Excel).

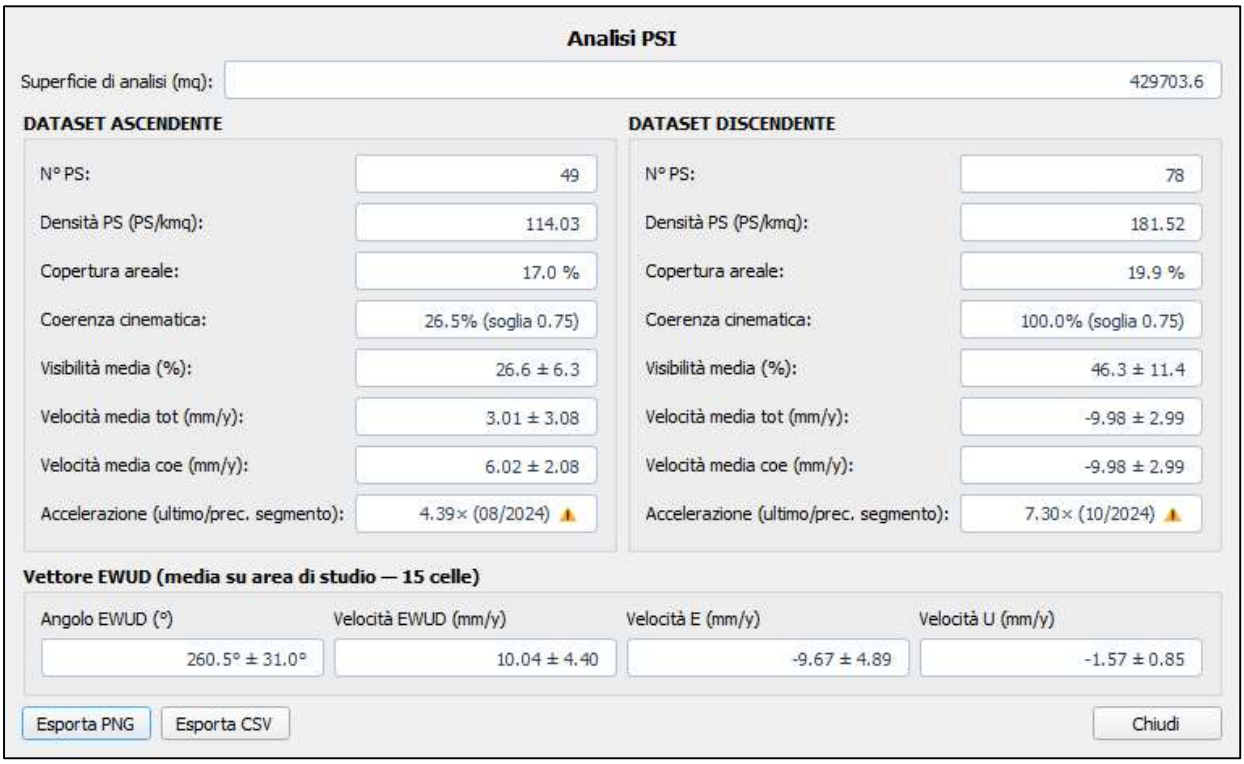


Fig. 11 — Results window of the InSAR Report module: analysis area, indicators for the ascending and descending datasets (N PS, density, areal coverage, kinematic coherence with threshold, visibility, velocity and acceleration with standard deviation/details) and mean EWUD vector over the study area.

10. Typical Workflow

The following is a complete workflow showing how the six modules fit together in a PSI analysis.

1. Data loading (Load) — Use Carica PS da File to select the GeoPackage, shapefile or GDB containing the PS data. Select the polygons of the index layer to load the ascending and descending PS layers in the area of interest.

2. EWUD decomposition — Create the resampling grid with a cell size suited to the PS density. Run the decomposition with the satellite preset matching the dataset (Sentinel-1 EGMS for EGMS Italy data).
3. Morphological analysis (VIS) — Load the DEM of the area. Compute pc_mov for the PS layer of interest, to assess where the SAR geometry is more or less sensitive to the expected movements.
4. Time series analysis (TS) — Set the PS layer as active. Select the PS points of interest. Run the analyses in this order: Qualità del dato → Analisi automatica → Scomposizione STL → Analisi non lineare → Anomalie temporali → Confronto tra zone. In each plot (except Anomalie temporali) the “Carica PS coerenti in QGIS” button is available.
5. Delineation of deformation areas (Polygons) — Select the PS layer. Open the InSAR Polygons module and set the velocity threshold, buffer radius and validation criteria. Start the calculation: the polygon layer, with a legend classified by velocity class, is loaded automatically.
6. Summary report (Report) — Define the study area (drawing, layer or file). Select the ascending/descending PS layers and, if available, the VIS and EWUD layers already computed over a wider extent. Click "Genera report": the summary window can be exported as PNG or CSV for the technical report.

i Each module is self-contained and can be used independently of the others. There is no need to follow the full sequence if data from some of the steps are already available.